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Estimating the reduction of urban PM10 concentrations by trees within an environmental information system for planners

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1. Abstract & Scope

Urban trees remove particulate matter (PM • €) from the air, but the magnitude of this effect is highly variable. We modeled PM • € deposition velocities (Vd) across the West Midlands, UK, using different planting scenarios (coniferous vs. deciduous trees) and meteorological datasets. The model integrated canopy structure, wind speed, and tree density. Coniferous species (Pinus nigra, Cupressocyparis leylandii) exhibited deposition velocities up to three times higher than deciduous species. Increasing canopy cover by 25% was estimated to reduce urban PM • € concentrations by 2% to 10%, leading to significant health benefits.

2. Methodology & Experimental Parameters

Atmospheric deposition velocity modeling (Vd) integrated into a Geographic Information System (GIS) and coupled with urban boundary layer models.

3. Core Findings & Data Synthesis

Coniferous canopies are highly efficient due to year-round leaf retention and high aerodynamic roughness. Broadleaved species are limited by leaf shedding and planar geometries.

4. Strategic Relevance to JU and Kolkata Planning

Provides the mathematical basis for Jadavpur University's green belt plans, justifying high-density canopy planting near campus borders.